Chapter - 8

Functions & Recursions

A function is a group of statements performing a specific task.

When a program gets bigger in size and its complexity grows, its difficult for a programmer to track on which piece of code is doing what!

A function can be reused by the programmer in a given program any number of.

Example and Syntax of a function -> The syntax of a function looks as follow.

```
def func1():
    Print("Hello")
```

This function can be called any number of times, anywhere in the program.

Function Call -> Whenever we want to call a function we put the name of the function followed. by paranthesis as follows:

func1() ⇒ This is called function call

Function definition -> The part containing the exact set of instructions which are executed during the function call.

Quick Quiz: Write a program to great a user with "Good day" using functions.

Types of functions in Python

1> Built in functions -> Already Present in Python.

2> User defined functions -> Defined by the user.

Examples of built in function includes len(), print() range() etc.
The func1() function we defined is an example of User defined function.

Functions with arguments

A functions can accept some values it can work with. We can put these values in the paranthesis.
A function can also return values as shown below

```
def greet(name):
        gr = "Hello" + name
        return gr

a = greet("Harry")
```

"Harry" is passed to greet in name

a will now contain "Hello Harry"

Default Parameter Value

We can have a value as default argument in a function.
If we specify name = "stranger" in the line contain def, this value is used when no argument is passed. For example:

```
def greet()
```

→ Name will be "stranger" in function body.

```
greet("Harry")
```

→ Name will be "Harry" in function body (passed)

```
def greet(name="Stranger"):
        # function body
```

Recursion

Recursion is a function which calls itself. It used to directly use a mathematical form as a function.

For example:

Factorial(n) = n × factorial(n-1)

This function can be defined as follows:

```
def factorial(n):
    if i==0 or i==1:  → Base condition which doesn't call
        return 1        the function any further
    else:
        return n * factorial(n-1) → Function calling itself
```

This work as follows:

Factorial(4) → [Function called]
↓
4 × Factoria(3)
4 × [3 × factorial(2)]
4 × 3 × [2 × factorial(1)]
4 × 3 × 2 × [1] × [factorial returned]

The programmers need to be externally careful while working with recursion to ensure that the function doesn't initially keep calling itself.
Recursion is sometimes the most direct way to code an algorithm.

Project 1: Snake, Water, Gun Game

We all have played snake, water gun game in our childhood. If you haven't, google the rules of this game and write a Python program capable of playing this game with the user.

Chapter 8 - Practice Set

1. Write a program using function to find greatest of three numbers.

2. Write a python program using function to convert celsius to fahrenheit.

3. How do you prevent a python print() func to print a new line at the end.

4. Write a recursive function to calculate the sum of first n natural numbers.

5. Write a python function to print first n of the following pattern:

```
* * *
* *
*
```

→ For n = 3

6. Write a python function which converts inches to cms

7. Write a python function to remove a given word from a list and step it at the same time.

8. Write a python function to print multiplication table of a given number.

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EXPLORER

CHAPTER-8_FUNCTIONS_&_RECURSIONS.PY

- 1.py
- 2026-03-25_17-55.py
- 2026-03-25_17-56.py
- default_arguments.py
- func_arguments.py
- functions.py
- Practice_set.ipynb
- quick_quiz.py
- recursion.py
- screen.jpg

OUTLINE

TIMELINE

default_arguments.py

```python
def goodday(name,ending="Thank You!"):
    print(f"Hello {name}, today is a good day!")
    print(ending)

goodday("Virendra", "Have a nice day!")
```

func_arguments.py

```python
def GoodDay(name):
    print(f"Hello {name}, today is a good day!")
    return "Done"

GoodDay("Virendra") # function call with argument
a=GoodDay("Virendra")
print(a)
```

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EXPLORER

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- default_arguments.py
- func_arguments.py
- functions.py
- Practice_set.ipynb
- quick_quiz.py
- recursion.py
- screen.jpg

OUTLINE

TIMELINE

functions.py > ...

```python
def avg():
    a=int(input("Enter the first number: "))
    b=int(input("Enter the second number: "))
    c=int(input("Enter the third number: "))
    average=(a+b+c)/3
    print("The average of the three numbers is: ", average

avg() # Calling the function to execute the code and c
```

Practice_set.ipynb

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[1]

```python
#Q-1
def greatest_num(n1, n2, n3):
    if n1>=n2 and n1>=n3:
        return f"{n1} is greatest"
    elif n2>=n1 and n2>=n3:
        return f"{n2} is greatest"
    else:
        return f"{n3} is greatest"
greatest_num(10, 20, 15)
```

Python

[3]

```python
#Q-2
def celcius_to_fahrenheit(celcius):
    fahrenheit=(celcius * 9/5) + 32
    return f"{celcius} degree celcius is equal to {fa

celcius_to_fahrenheit(3)
```

Python

[4]

```python
#Q-3
print("hello world")
print("hello world")
print("hello world",end="")
```

Python

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